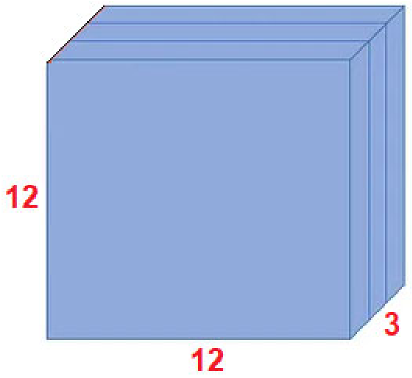
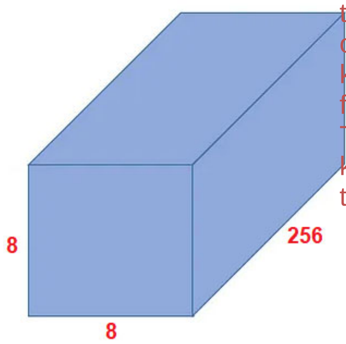
SC4172 Tiny Machine Machine Learning
Tutorial 5 –Depthwise Separable Convolution

A 12x12 color image is to be convoluted using Depthwise Separable (DWS) Convolution to produce a 8x8 x 256 output tensor (using kernel of 5x5 size for DWS)





2 stage, Depthwise convolution, then pointwise convolution. Instead of using kernel of 5x5x3 and 256 kernels straight away.Use 5x5x1 first, then forms 8x8x3 tensor output. Then use 1x1x3 kernel with 256 kernels then forms a 8x8x256 output tensor.

a. Describe how this can be achieved.
b. Calculate the total number of MAC operations.

   (5x5x3x8x8)+(3x1x1x8x8x256[no. of kernels])
   DEPTHWISE + POINTWISE convolution

c. How many parameters will be generated in this layer of the model?

   1st stage uses 3 kernels, weights = 5x5x1x3
   2nd stage uses 256 kernels, weights = 1x1x3x256
   total weights = 75+768 = 843